

ASTR/PTYS 578: Assignment 4 Part A,B & C

Part A& B DUE : Feb 11th 2026 by **NOON**

FORMAT: Word doc

FILENAME: LastName_Assignment4_PartAB

Part A: Figure and Caption

Construct a **figure** and **caption** that outlines a strategy for a solution to a problem that you will use HST/JWST/Theoretical Model to solve.

i.e. the figure and caption convey: Target + Strategy = Solution.

Figure Details:

- The figure can be a diagram or sketch – you can design it by hand/power point etc. It does not have to be the final polished version that you will use in the proposal.
- You can modify a figure from a published paper – make sure to cite the paper.
- The figure can have multiple panels.
- Somewhere in the figure itself should be the words: “This proposal” to highlight the strategy being proposed.

Figure caption must follow the following format:

- First sentence: bold faced takeaway that highlights the broader impact of the program. Effectively states the outcome of the program.
- Second(+) sentence(s): Description of what is plotted
- Following the description is a statement of the problem related to the target, which will be solved by the plot.
- Last sentence(s) is the Goal. Must include : “This program will” ... or “We propose to” AND the importance of the solution to the subfield. Underline the goal.

Part B: Talk Abstract about the Figure

Create a talk abstract centered on your figure. Include this in the doc after the Part A.

- 1) Determine a talk title that is connected to “the problem” that is solved by your figure from Part A. Follow the “[Big Picture]: [Direct Problem]” format.
- 2) Construct a talk abstract as follows:
 - One sentence : one big picture Fact
 - One sentence : Importance of the Fact.
 - One sentence: Problem – this is **set up** by the first two sentences.
 - One Sentence: Goal: “In this talk I will discuss/demonstrate a **strategy** to solve **the problem**”. [This maps on to “we propose”]
 - If you don’t know the solution yet, that’s ok. you can write one idea down for now and we can build from there in class.
 - These Sites can help you figure out what you want to propose for:
 - JWST: <https://webb.nasa.gov/content/science/index.html>
 - HST: <https://www.stsci.edu/hst/about/key-science-themes>
 - One Sentence: Urgency/Importance to subfield. Why now? Why this facility/funding call?
 - One Sentence: Future Outlook/Broader Impact. End with a sentence about the importance of solving “the Problem” broadly to the field.
- Highlight each of the above components in your abstract in a **different color**.
- Abstract must be no longer than **20 lines of 85 characters of text** (including spaces). In other words max 1700 characters including white spaces.

Part C: Presentation

Create a presentation slide deck based on your figure and abstract, and present the slides in a short (7 min) talk to the class. The grade is based on the slide quality and presentation.

FORMAT: Oral presentation on schedule below & Upload a PDF of your presentation slides.

Format for Upload: "LastName_Assignment4PartC.pdf"

Deadlines:

- Upload your slides by March 5th at 5 PM on D2L.
- Presentation Schedule:
 - Feb 19th: Nick S., Carter M., Kayla S., Kylie H., Rafa B.
 - Feb 24nd: Maddy C., Ruby F., Devin H., Lily R., Tyler R.,
 - March 3rd: Aaron G, Alena C., Emily M, Madi V., Yichen L
 - March 5th: Lipika C., Cole M., Neev S., Chen-Yu C., Sophie C.

Duration: 7 minutes + 3 min discussion

Format/Instructions: Turn your abstract into a 7 slide presentation (with relevant images):

Slide 1) Title Slide – spend 1 minute setting the stage!

Include: title, your name, one picture that is evocative

Slide 2) (1-2 slides max) Fact – this should correspond to the first sentences of your abstract. Set the background needed to understand the context for the problem

Slide 3) Importance of fact. (WHY does the topic MATTER – why should the audience care?)

Slide 4) The Target – what are you proposing to study/model?

Slide 5) The Problem – convey urgency in solving this *now*

Slide 6) Proposed Solution & Importance/Urgency (This will turn into the “we propose” e.g. Target + Proposed Data = Solution). This is where the Figure you created in Part A should be presented.

Slide 7) Broader Impacts/Future Outlooks: What future observatories/programs will benefit from your proposal?

- Each slide must have a heading that states the *takeaway* point for the slide.
- You must explain the axes for each figure.
- When asked questions, you must repeat the question back to the asker.

Grading Criteria for the entirety of Assignment 4: (20% of final grade)

- Part A: Follow the Structure Outlined Above for the Figure and Caption (/10)
- Part B: Follow the Structure Outlined Above for the Abstract (/10)
- Part C: (/20)
 - o Follow the Structure Outlined Above for the Presentation (/10)
 - o Quality of Presentation – clarity of slides & oral presentation (/10)Participation (asking **at least 3** questions to others)